RAHUL A

PORTFOLIO & PROOF OF WORK DOCUMENTATION

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Executive Profile: B.Tech in Digital Transformation at **Atria University, Bengaluru** (2024–2028). Currently an **AI & Full Stack Engineering Intern** at **OpenAlgon / Decembay Technologies** (May 2026 – Present), where I have shipped **9 live production applications** across education, healthcare, hospitality, agriculture, and enterprise domains. Primary career focus in **Cybersecurity** (VAPT, Ethical Hacking, OWASP Top 10, Cryptographic Ledgers, and Secure API Architecture), paired with demonstrable proof-of-work in edge full-stack engineering and zero-leakage machine learning pipelines.

1. PROFESSIONAL INTERNSHIP — OPENALGON | DECEMBAY TECHNOLOGIES (9 SHIPPED APPS)

Role: AI & Full Stack Engineering Intern | **Duration:** May 2026 — Present | **Companies:** OpenAlgon / Decembay Technologies

Shipped 9 live full-stack web and cross-platform applications in production environments:

1. RankPrediction — AI Counseling Intelligence Platform [Live: https://rankprediction.com | GitHub: rahul200618/Rankpredictor]

Stack: React 18 · TypeScript · Vite · Supabase · PostgreSQL · Firebase Auth · Recharts · Vercel

- **Rank Predictor Engine:** Formulated scoring logic calibrated against thousands of historical KCET & COMEDK cutoff ranks across categories and engineering branches.
- **AI Counselor & Trends:** Integrated option-entry recommendations, round tracking, and cutoff visualization via Recharts with phone Firebase Auth + reCAPTCHA.

2. Shree Udupi Aramane — Restaurant Operations Monolith & POS [Live: https://shreeudupiaramane.com/]

Stack: Next.js 16 (App Router) · React 19 · TypeScript · Express · Prisma · Supabase · BullMQ · Socket.IO · Cashfree

- **4 Operational Portals:** Serves Customer Storefront, Kitchen Display System (KDS), Delivery Valet, and Admin Dashboard on a unified monolith port 5000.
- **FSM & Financial Precision:** Role-gated order state machine (Placed ➔ Delivered), Socket.IO live sync, BullMQ transactional outbox, and integer paise accounting.

3. Abhyas Nepal — Medical Entrance Examination Platform [Live: https://abhyas.pro/]

Stack: TanStack Start · React 19 · TypeScript · Tailwind CSS · Supabase · Vitest · KaTeX

- **CEE Exam Engine:** Timed full-length model exams and chapter MCQs with KaTeX formula rendering, accuracy analytics, and All-Nepal leaderboards.
- **Integrity & Testing:** Phone OTP authentication, rate limiting, anti-cheat test attempts, and automated component/logic test suites with Vitest.

4. Nepal CEE College Predictor — Cloudflare Edge SSR [Live: https://nepal-college-predictor.pages.dev/ | GitHub: rahul200618/nepal-college-predictor]

Stack: TanStack Start · React 19 · Tailwind CSS v4 · Supabase · Nitro · Cloudflare Pages

- **Admission Prediction Engine:** Marks-to-rank estimation algorithm evaluating 16+ medical bachelor programs (MBBS, BDS, Nursing) across all 14 official MEC reservation quotas.
- **Edge SSR:** Deployed on Cloudflare Pages via Nitro server engine for sub-50ms TTFB.

5. Agro ERP (Field Flow) — Multi-Platform Enterprise ERP [Live: https://agro.openalgon.com]

Stack: TanStack Start · React 19 · TypeScript · Drizzle ORM · Neon DB · Capacitor · Tauri · Three.js · Razorpay

- **Web + Android + Desktop:** Multi-platform deployment from a single codebase (Capacitor for Android, Tauri for Desktop).
- **Multi-Shop Isolation & POS:** 100% tenant data isolation, live camera QR scanning (html5-qrcode), Razorpay payments, and Three.js 3D product visualizations.

6. AgriArchives — International Monthly Agriculture E-Magazine [Live: https://agriarchives.in | GitHub: rahul200618/agriarchive]

Stack: React 18 · TypeScript · Tailwind CSS · shadcn/ui · Supabase · PostgreSQL · Fuse.js · Vercel

- **Auth-Gated Admin CMS:** Secure publishing pipeline for editors to upload and archive monthly journal editions with Supabase PDF storage and Fuse.js sub-ms fuzzy search.

7. AgriCatalogues — Research Journal & Article Portal [Live: https://agricatalogues.in]

Stack: React · TypeScript · Vite · Tailwind CSS · shadcn/ui · Supabase · React Router DOM

- **Searchable Research Catalog:** Dynamic category filtering, type-safe schema for author profiles, and digital author submission workflows.

8. Krishikuta — Agri-Education & Professional DPR Consulting [Live: https://krishikuta.com | GitHub: rahul200618/krushikuta]

Stack: React 19 · TypeScript · TanStack Router · TanStack Query · Tailwind CSS · Supabase · Framer Motion

- **Exam Coaching & DPR Funnels:** Prep modules for ICAR/AO/K-CET and client intake for commercial agricultural Detailed Project Reports (dairy, poultry, goat).

9. Bluedot Immigrations — Immigration Portal & Scheduling System [Live: https://bluedotimmigrations.com]

Stack: TanStack Start · React 19 · TypeScript · Tailwind CSS v4 · shadcn/ui · Supabase · Cloudflare Workers

- **Consultation Engine:** Consultation scheduling and inquiries with an authenticated Admin dashboard visualizing lead conversion metrics via Recharts on Cloudflare Workers.

2. TECHNICAL PROOF OF WORK — PIPELINES & ARCHITECTURAL DECISIONS

Evaluated by technical hiring panels to verify genuine pipeline engineering, zero data leakage, and security rationale:

A. End-to-End Supervised ML Pipeline (Wine Quality) — AUC-ROC: 0.816 [Repo: rahul200618/wine-quality-ml]

Stack: Python · Scikit-learn · Pandas · NumPy · Matplotlib · EDA

- **Why Stratified Splitting & Train-Only Scaling?** Target classes were imbalanced (63% good vs 37% poor). A 60/20/20 stratified split ensured equal class distributions across splits. StandardScaler was fit *strictly on training data* to eliminate test leakage. Evaluated Linear/Logistic Regression and tuned k-NN Regressor/Classifier across 6,497 samples.
- **Metrics Achieved:** Binary classification AUC-ROC of 0.816; k-NN hyperparameter tuning minimized validation RMSE on continuous scale (3–9).

B. Cryptographic Digital Cash & Double-Spend Attack Simulation [Repo: rahul200618/Minimal-Digital-Cash-System-Cryptography]

Stack: Python · ECDSA (secp256k1) · SHA-256 Hash Chaining · Cryptographic Verification

- **Architecture & Attack Suite:** Engineered an append-only, tamper-evident ledger where blocks cover previous block hashes. Created secp256k1 ECDSA wallets with deterministic key-pair derivation and nonce-based transaction ordering.
- **Automated Attack Simulation:** Wrote attack_simulation.py running automated exploits against 6 attack vectors: transaction replay, double-spend attempts, payload tampering, forged signatures, balance overdrafts, and chain tampering.

C. Symbio Insight — AI Bioinformatics FASTA Sequence Analyzer [Repo: rahul200618/symbio-insight-app]

Stack: Python · JavaScript · Google Gemini API · Bio-parser · Render Deploy

- **Design:** Full-stack platform parsing FASTA DNA/RNA biological sequence files, extracting GC-content, Open Reading Frames (ORFs), and streaming AI-driven biological function summaries via Google Gemini API.
- **Deliverables:** Automated PDF/HTML report generator and production live app at symbio-insight-app-2.onrender.com.

D. Formula Bharat 2026 — RevX Atria (E17 Team) | EV Engineering [Atria Racing Team]

Stack: EV Powertrain Integration · BMS Architecture · High/Low Voltage Harness

- **Core Engineering Role:** Member of Atria University Racing Team (2025–Jan 2026) designing an electric race vehicle under Formula Bharat regulations. Hands-on exposure to EV drivetrain integration, battery pack topology, and competition telemetry.
- **Impact:** Coordinated cross-functional mechanical, electrical, and firmware interfaces for vehicle scrutineering compliance.

3. TECHNICAL SKILLS & DOMAIN MATRIX

Domain	Tools, Frameworks & Core Concepts
Cybersecurity (Primary Focus)	VAPT, OWASP Top 10, Ethical Hacking, Kali Linux, Network Security, Secure API Design, JWT Auth & Refresh Rotation, Cryptographic Signatures (ECDSA), Tamper-evident Hash Chains, Threat Modeling
Full-Stack Engineering	Next.js 16, React 18/19, TypeScript, TanStack Start/Router/Query, Node.js, Express, Tailwind CSS v4, shadcn/ui, Supabase (PostgreSQL, RLS), Prisma ORM, Drizzle ORM, Neon DB, BullMQ, Socket.IO, Cashfree, Cloudflare Workers
Machine Learning & Data	Supervised & Unsupervised Learning, Scikit-learn, Pandas, NumPy, StandardScaler (Zero Leakage), Stratified Splits, Power BI, Advanced SQL, EDA, Anomaly Detection, PCA Dimensionality Reduction
Cross-Platform & AI	Capacitor (Android/iOS), Tauri (Desktop), Google Gemini API, LLM Integration, AI Agents/MCP Concepts, Three.js 3D WebGL Visualization, Razorpay & Cashfree Payment Gateways, QR Code Scanning
Languages	Python, JavaScript, TypeScript, SQL, HTML5, CSS3 | English (Fluent), Kannada (Native), Hindi (Skillful)

4. EDUCATION & INDUSTRY SIMULATIONS

B.Tech in Digital Transformation (2024–2028) Atria University, Bengaluru	Primary focus on Cybersecurity, Data Science, and Full-Stack Systems. Active member of RevX Atria Racing (Formula Bharat).
Secondary (10th) & Senior Secondary (12th)	Kendriya Vidyalaya Malleswaram, Bengaluru (10th) & Senior Secondary STEM (12th Graduated 2024).
Cybersecurity Analyst Job Simulation — Tata & Forage	Simulated enterprise security operations: threat vectors, vulnerability assessment, incident response.
Data Visualisation Simulation — Tata & Forage	Business intelligence reporting, KPI dashboarding, and stakeholder data presentations.